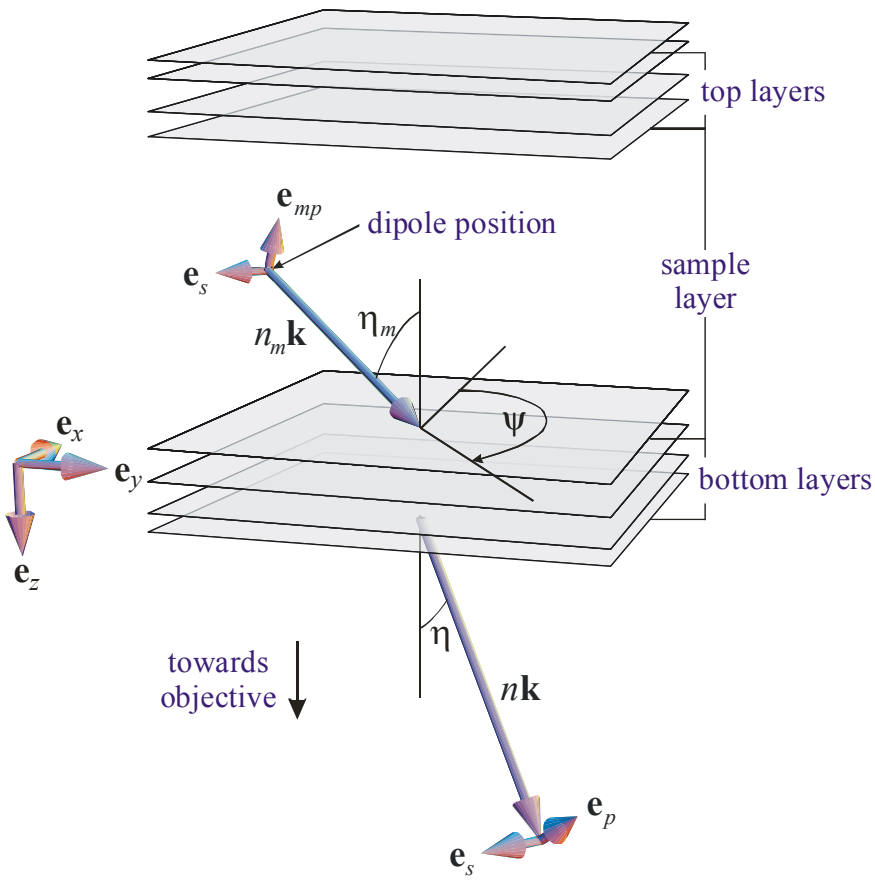
Program LifetimeL.m

Syntax:

```
[lvd,lvu,lpd,lpu,qvd,qvu,qpd,qpu] = LifetimeL(z,n1,n,n2,d1,d,d2)
```

Description:

LifetimeL calculates the lifetime and quenching of an oscillating dipole at a distance z from the bottom interface of a layer with refractive index n and thickness d that is sandwiched between two stacks of layers with refractive indices $n_{1,1}$ through $n_{1,k1}$ and thickness values $d_{1,1}$ through $d_{1,k1-1}$ (bottom stack, from bottom to top) and refractive indices $n_{2,1}$ through $n_{2,k2}$ and thickness values $d_{2,1}$ through $d_{2,k2-1}$ (top stack, from bottom to top).



Input variables: (all length values in units of emission wavelength over 2π)

z – molecule's distance from the bottom of its layer

n1 – vector of refractive indices of the stack below the molecule's layer

n – refractive index of the molecule's layer

n2 – vector of refractive indices of the stack above the molecule's layer

d1 – vector of layer thickness values of the stack below the molecule's layer

(length(d1) = length(n1)-1)

d – thickness of molecule's layer

d2 – vector of layer thickness values of the stack above the molecule's layer

(length(d2) = length(n2)-1)

Output variables (all emission powers are in such units so that a free dipole within a homogeneous medium with refractive index *n* has an emission power of 4/3·*n*):

lvd – Emission power of vertically oriented dipole (perpendicular to interfaces) into infinite bottom half-space;

lvu – Emission power of vertically oriented dipole (perpendicular to interfaces) into infinite top half-space;

lpd – Emission power of parallel oriented dipole (parallel to interfaces) into infinite bottom half-space;

lpu – Emission power of parallel oriented dipole (parallel to interfaces) into infinite top half-space;

qvd – Total emission power of vertically oriented dipole (perpendicular to interfaces) towards the bottom stack of layers;

qvu – Total emission power of vertically oriented dipole (perpendicular to interfaces) towards the top stack of layers;

qpd – Total emission power of parallel oriented dipole (parallel to interfaces) towards the bottom stack of layers;

qpu – Total emission power of parallel oriented dipole (parallel to interfaces) towards the top stack of layers;

Example 1:

One would like to calculate the lifetime of a molecule within a thin polymer film with refractive index 1.4 sandwiched between a glass substrate (bottom) with refractive index 1.51 and air (top) with refractive index of 1.0. The thickness of the polymer film is 4 μm, the calculations should be performed for varying vertical molecule position between 0 and 4 μm. Emission wavelength is 570 nm. Thus, the input parameters are

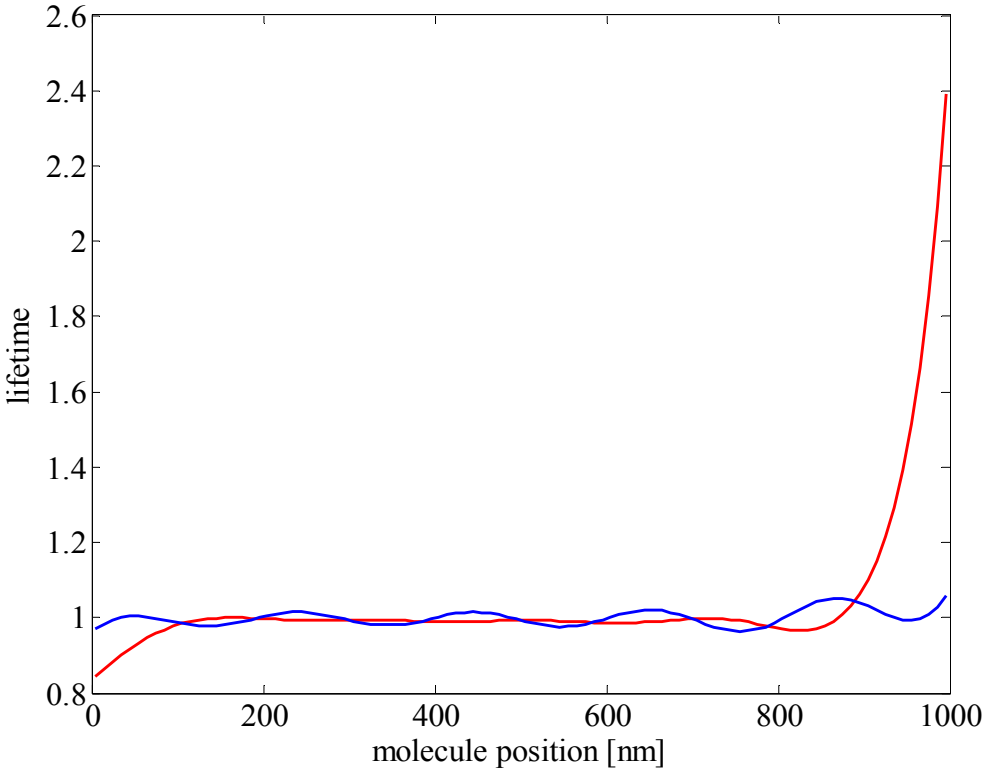
```
lam = 570;
n1 = 1.51; n = 1.4; n2 = 1.0;
d1 = []; d = 1000/lam*2*pi; d2 = [];
z = (0.5:100)/100*1000/lam*2*pi;
```

and the calculation is done by

```
[lvd,lvu,lpd,lpu] = LifetimeL(z,n1,n,n2,d1,d,d2);
```

The lifetime (with respect to a dipole within an infinite polymer with refractive index $n = 1.4$) is then plotted by

```
plot(z/2/pi*lam,4/3*n./(lvu+lvd),z/2/pi*lam,4/3*n./(lpu+lpd));
ylabel('lifetime');
xlabel('molecule position [nm]')
```



Red shows the vertical and blue the parallel dipole orientation.

Example 2:

One would like to calculate the lifetime and fluorescence quenching of a molecule at 570 nm emission that is situated above a 20 nm gold layer on glass in air. The complex-valued refractive index of gold at 570 nm is 0.3257 + 2.5792·*i*.

Thus, one has

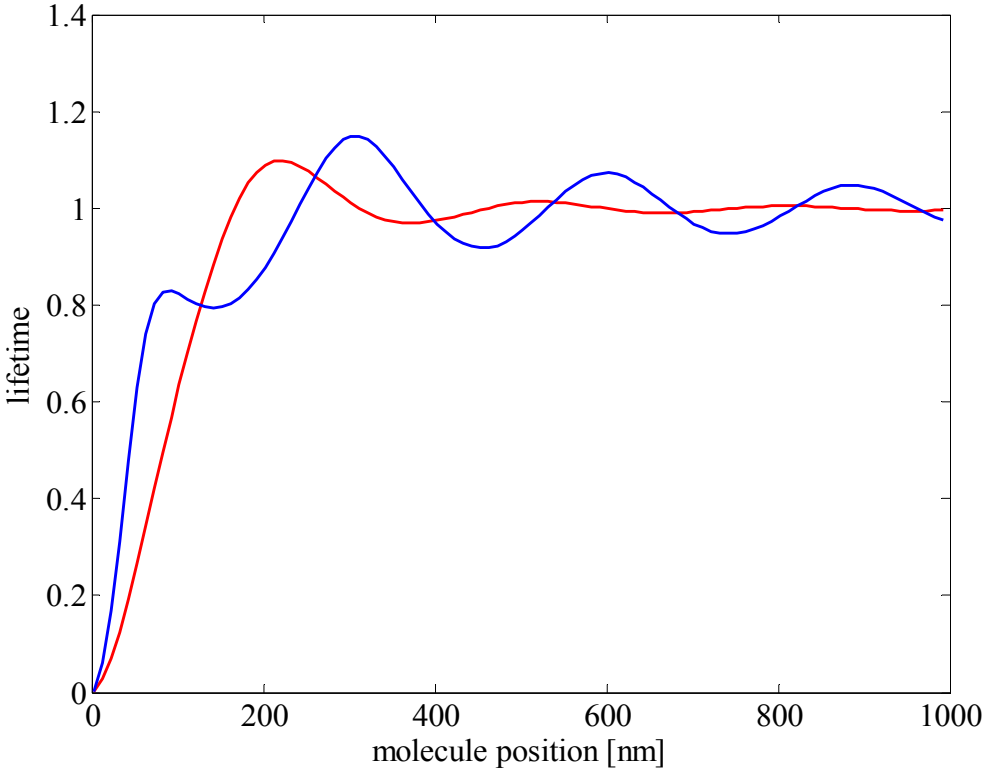
```
lam = 570;
n1 = [1.51, 0.3257 + 2.5792i]; n = 1.0; n2 = 1.0;
d1 = 20/lam*2*pi; d = 1000/lam*2*pi; d2 = [];
z = (0.25:100)/100*1000/lam*2*pi;
```

and the calculation is done by

```
[lvd,lvu,lpd,lpu,qvd,qvu,qpd,qpu] = LifetimeL(z,n1,n,n2,d1,d,d2);
```

The lifetime (with respect to a free dipole within air) is plotted by

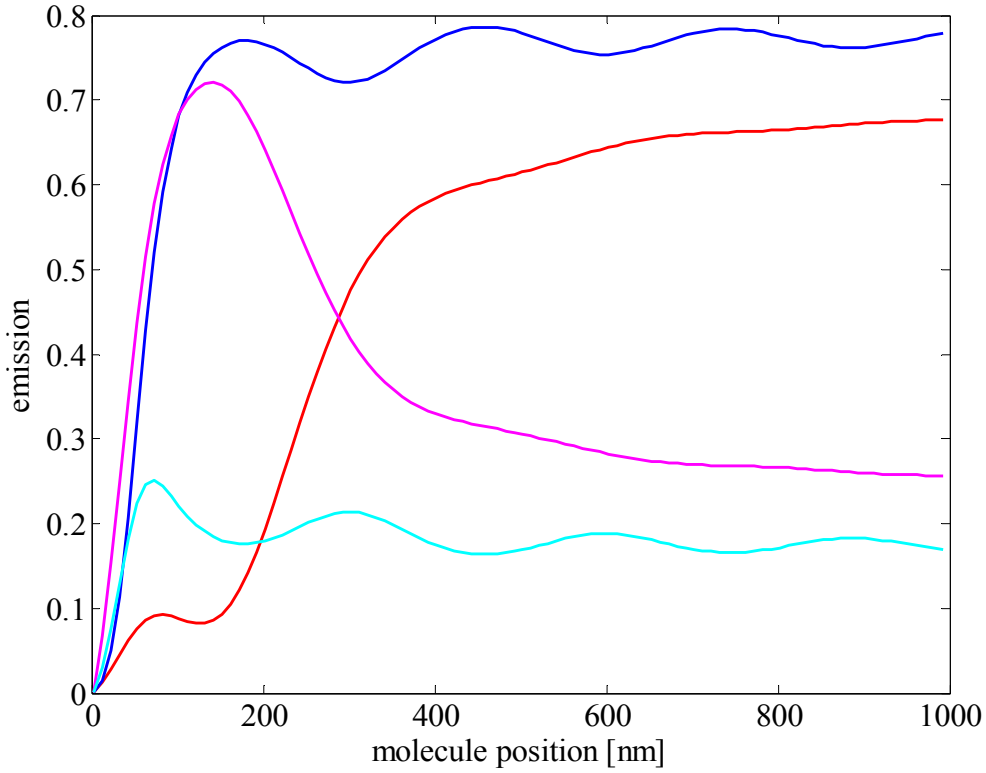
```
plot(z/2/pi*lam,4/3*n./(qvu+qvd),z/2/pi*lam,4/3*n./(qpu+qpd));
ylabel('lifetime');
xlabel('molecule position [nm]')
```



Red shows the vertical and blue the parallel dipole orientation.

The amount of fluorescence emitted into the top half space is shown by

```
plot(z/2/pi*lam,lvu./(qvu+qvd),z/2/pi*lam,lpu./(qpu+qpd),
z/2/pi*lam,lvd./(qvu+qvd),z/2/pi*lam,lpd./(qpu+qpd));
ylabel('top emission');
xlabel('molecule position [nm]')
```



Here, red and blue show the emission into air for the vertical and parallel dipole orientation, respectively, and magenta and cyan show the emission into glass for the vertical and parallel dipole orientation, respectively.